```
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# div : B
# Roll No :31
# PRN : 72332154L
# DMV PRATICAL NO 6
```

```
import kagglehub
import pandas as pd
import matplotlib.pyplot as plt
import os

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
path = kagglehub.dataset_download("abuhumzakhan/store-data")

print("Dataset Path:", path)
print("Files in Dataset Folder:")
print(os.listdir(path))
```

```
Downloading to /root/.cache/kagglehub/datasets/abuhumzakhan/store-data/1.archive...
100%|██████████| 5.21M/5.21M [00:00<00:00, 104MB/s]Extracting files...

Dataset Path: /root/.cache/kagglehub/datasets/abuhumzakhan/store-data/versions/1
Files in Dataset Folder:
['store_sales_data (2).csv']
```

```
files = os.listdir(path)

csv_file = [file for file in files if file.endswith(".csv")][0]
file_path = os.path.join(path, csv_file)

df = pd.read_csv(file_path)

print("Dataset loaded successfully!")
display(df.head())
```

```
Dataset loaded successfully!
```

	Customer ID	Customer Name	Last Name	Date of Birth	Sales	Year	Outlet Type	City Type	Category of Goods	Region	...	Ship Date	Ship Mode	State	Postal Code	Product ID	Sub-Category	Product Name
0	CUST000001	Curtis	Krause	1973-11-29	3344.69	2023	Large	Tier 1	Fast Food	East	...	2020-04-18	Same Day	West Bengal	700001	PROD000001	Burgers	Burgers - 718
1	CUST000002	Jeremy	Foster	1987-08-30	37094.41	2022	Medium	Tier 2	Fast Food	East	...	2021-01-11	Standard Class	Punjab	160001	PROD000002	Pizzas	Pizzas - 519
2	CUST000003	Ariel	Mccann	1980-12-15	35501.08	2023	Large	Village	Electric Appliances	South	...	2020-08-16	Second Class	Gujarat	380001	PROD000003	Fans	Fans - 395
3	CUST000004	Colleen	Maynard	1989-05-23	8969.33	2021	Medium	Village	Dairy Products	East	...	2021-10-16	First Class	West Bengal	700001	PROD000004	Milk	Milk - 798
4	CUST000005	Gina	Holden	1957-12-31	35530.44	2021	Small	Village	Sessional Fruits & Vegetables	West	...	2019-01-25	Standard Class	Delhi	110001	PROD000005	Tomatoes	Tomatoes - 442

5 rows × 25 columns

```
print("Dataset Shape:", df.shape)
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

```
Dataset Shape: (100000, 25)
Number of Rows: 100000
Number of Columns: 25
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 25 columns):
 #   Column             Non-Null Count   Dtype  
---  ------             --------------   -----  
 0   Customer ID        100000 non-null  object 
 1   Customer Name      100000 non-null  object 
 2   Last Name          100000 non-null  object 
 3   Date of Birth      100000 non-null  object 
 4   Sales              100000 non-null  float64
 5   Year               100000 non-null  int64  
 6   Outlet Type        100000 non-null  object 
 7   City Type          100000 non-null  object 
 8   Category of Goods  100000 non-null  object 
 9   Region             100000 non-null  object 
 10  Country            100000 non-null  object 
 11  Segment            100000 non-null  object 
```

```
 12  Sales Date          100000 non-null  object
 13  Order ID            100000 non-null  object
 14  Order Date          100000 non-null  object
 15  Ship Date           100000 non-null  object
 16  Ship Mode           100000 non-null  object
 17  State               100000 non-null  object
 18  Postal Code         100000 non-null  int64
 19  Product ID          100000 non-null  object
 20  Sub-Category        100000 non-null  object
 21  Product Name        100000 non-null  object
 22  Quantity            100000 non-null  int64
 23  Discount            100000 non-null  float64
 24  Profit              100000 non-null  float64
dtypes: float64(3), int64(3), object(19)
memory usage: 19.1+ MB
```

```
display(df.describe())
```

	Sales	Year	Postal Code	Quantity	Discount	Profit
count	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000
mean	25084.410142	2020.999910	388985.180000	5.493760	0.251320	3755.305114
std	14403.187686	1.413757	184192.704685	2.876102	0.144324	2639.852035
min	100.570000	2019.000000	110001.000000	1.000000	0.000000	6.520000
25%	12618.030000	2020.000000	226001.000000	3.000000	0.130000	1651.105000
50%	25134.695000	2021.000000	380001.000000	5.000000	0.250000	3317.450000
75%	37575.850000	2022.000000	560001.000000	8.000000	0.380000	5363.845000
max	49999.890000	2023.000000	700001.000000	10.000000	0.500000	14563.670000

```
print("Missing Values in Each Column:")
print(df.isnull().sum())
```

```
Missing Values in Each Column:
Customer ID          0
Customer Name        0
Last Name            0
Date of Birth        0
Sales                0
Year                 0
Outlet Type          0
City Type            0
Category of Goods    0
Region               0
Country              0
Segment              0
```

```
Sales Date          0
Order ID            0
Order Date          0
Ship Date           0
Ship Mode           0
State               0
Postal Code         0
Product ID          0
Sub-Category        0
Product Name        0
Quantity            0
Discount            0
Profit              0
dtype: int64
```

```
print("Relevant Variables:")

display(df[
    [
        "Region",
        "Category of Goods",
        "Quantity",
        "Sales",
        "Order Date"
    ]
].head())
```

Relevant Variables:

	Region	Category of Goods	Quantity	Sales	Order Date
0	East	Fast Food	3	3344.69	2020-04-13
1	East	Fast Food	5	37094.41	2021-01-08
2	South	Electric Appliances	6	35501.08	2020-08-11
3	East	Dairy Products	4	8969.33	2021-10-10
4	West	Sessional Fruits & Vegetables	4	35530.44	2019-01-22

```
region_sales = df.groupby("Region")["Sales"].sum()

print("Total Sales by Region:")
print(region_sales)
```

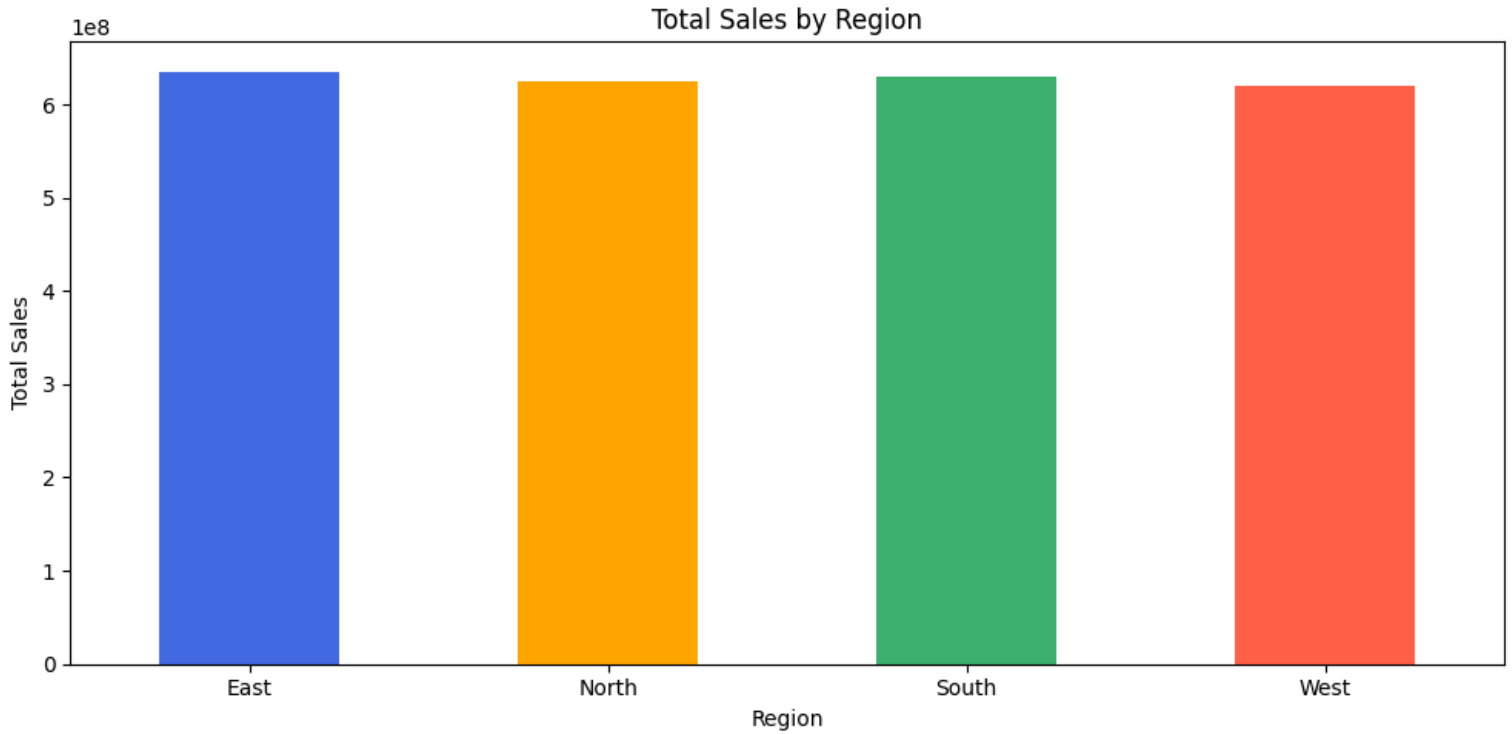
```
Total Sales by Region:
Region
East     6.353223e+08
North    6.244511e+08
South    6.290598e+08
```

```
West     6.196078e+08
Name: Sales, dtype: float64
```

```
plt.figure(figsize=(10, 5))

region_sales.plot(
    kind="bar",
    color=["royalblue", "orange", "mediumseagreen", "tomato"]
)

plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

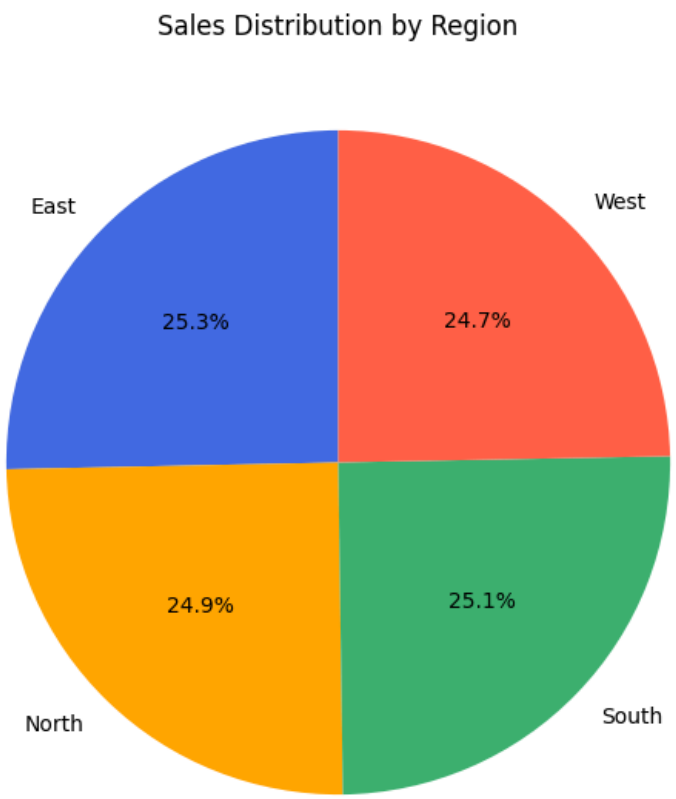



```
plt.figure(figsize=(7, 7))

region_sales.plot(
    kind="pie",
```

```
    autopct="%1.1f%%",
    startangle=90,
    colors=["royalblue", "orange", "mediumseagreen", "tomato"]
)

plt.title("Sales Distribution by Region")
plt.ylabel("")
plt.show()
```




```
top_regions = region_sales.sort_values(ascending=False)

print("Regions Ranked by Sales:")
print(top_regions)

print("\nTop 3 Performing Regions:")
print(top_regions.head(3))
```

```
Regions Ranked by Sales:
Region
East     6.353223e+08
South    6.290598e+08
North    6.244511e+08
West     6.196078e+08
Name: Sales, dtype: float64

Top 3 Performing Regions:
Region
East     6.353223e+08
South    6.290598e+08
North    6.244511e+08
Name: Sales, dtype: float64
```

```
region_category_sales = df.groupby(
    ["Region", "Category of Goods"]
)["Sales"].sum().unstack()

print("Sales by Region and Product Category:")
display(region_category_sales)
```

Sales by Region and Product Category:

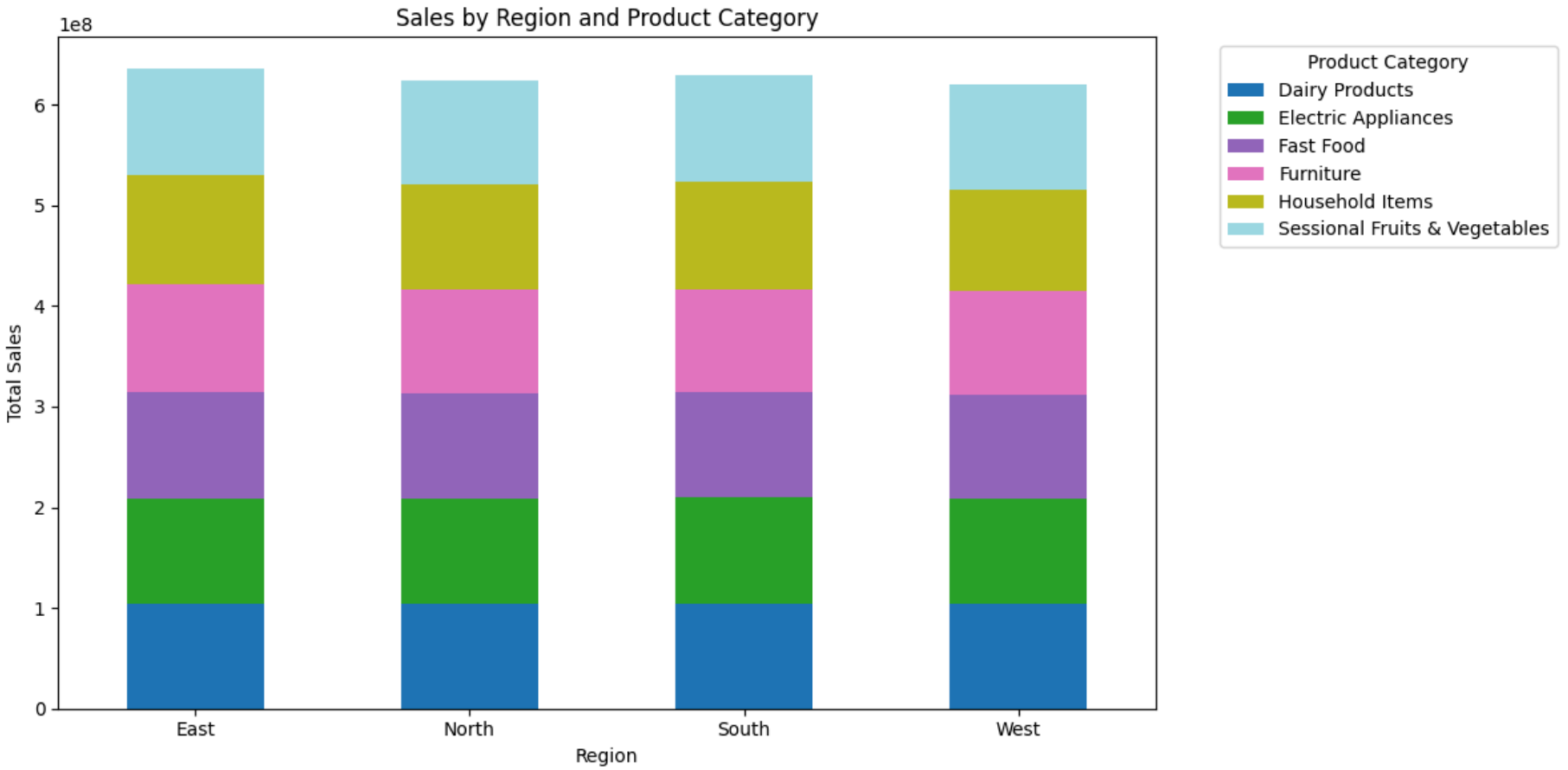
Category of Goods	Dairy Products	Electric Appliances	Fast Food	Furniture	Household Items	Sessional Fruits & Vegetables
Region						
East	1.043553e+08	1.040628e+08	1.065402e+08	1.064914e+08	1.082594e+08	1.056132e+08
North	1.046789e+08	1.044963e+08	1.044842e+08	1.025701e+08	1.039086e+08	1.043131e+08
South	1.044889e+08	1.057580e+08	1.041161e+08	1.016497e+08	1.067453e+08	1.063018e+08
West	1.044885e+08	1.045126e+08	1.027104e+08	1.033062e+08	1.008756e+08	1.037146e+08

Next steps: Generate code with `region_category_sales` | New interactive sheet

```
region_category_sales.plot(
    kind="bar",
    stacked=True,
    figsize=(12, 6),
    colormap="tab20"
)

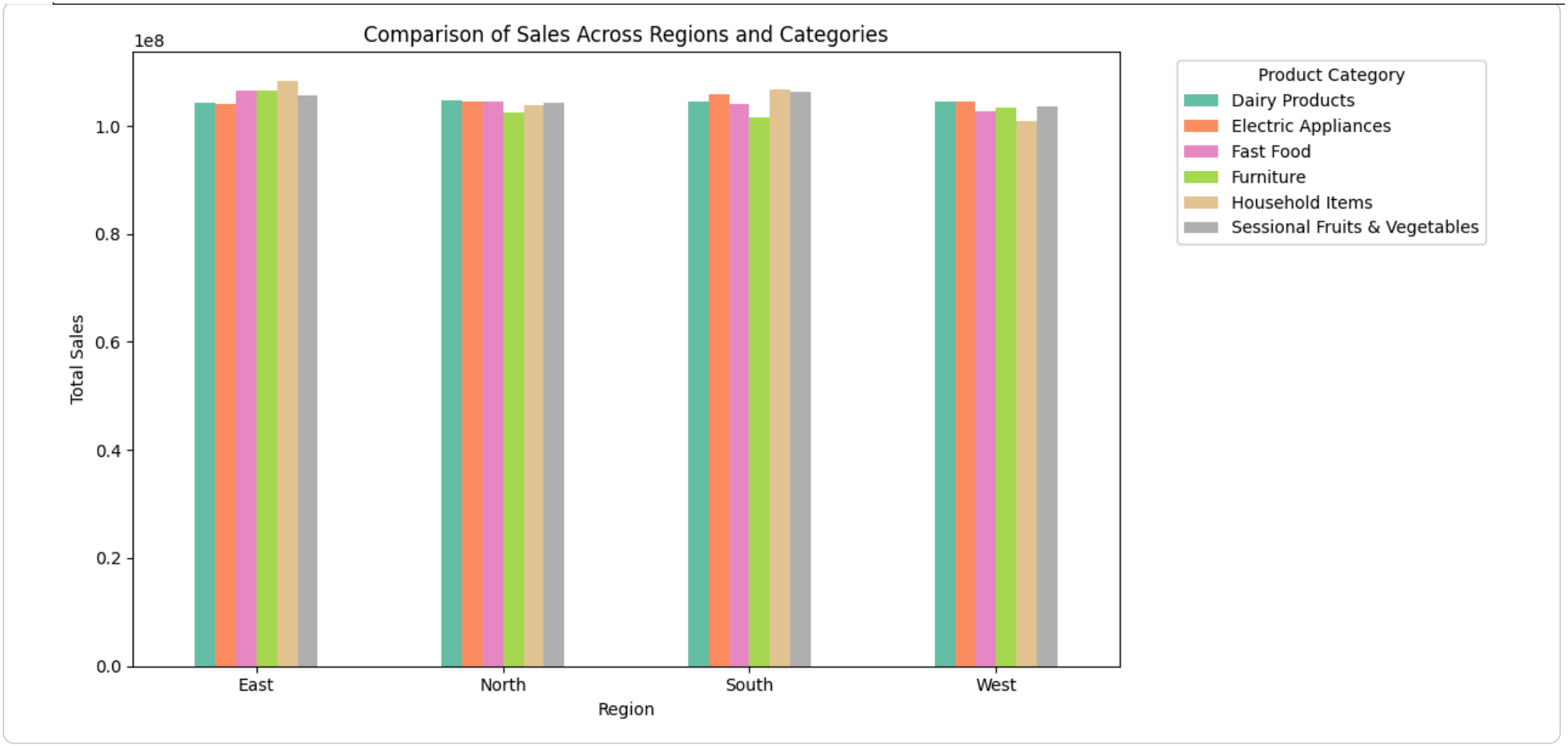
plt.title("Sales by Region and Product Category")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.legend(title="Product Category", bbox_to_anchor=(1.05, 1))
```

```
plt.tight_layout()
plt.show()
```




```
region_category_sales.plot(
    kind="bar",
    figsize=(12, 6),
    colormap="Set2"
)

plt.title("Comparison of Sales Across Regions and Categories")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.legend(title="Product Category", bbox_to_anchor=(1.05, 1))
plt.tight_layout()
plt.show()
```


Comparison of Sales Across Regions and Categories
1e8
1.0
0.8
0.6
0.4
0.2
0.0
Total Sales
East
North
South
West
Region
Product Category
Dairy Products
Electric Appliances
Fast Food
Furniture
Household Items
Sessional Fruits & Vegetables